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**Department of Electronics and Communication
Engineering**

A Literature review on Minor Project-1 [22URMK506] titled

**“AUTOMATIC MEDICINE
DISPENSER”**

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LITERATURE REVIEW

1. Background and Need for Automated Medication Systems

- Medication adherence is a major challenge in healthcare, especially among elderly patients, chronically ill individuals, and home-care patients.
- Traditional medicine administration relies heavily on manual supervision by nurses or caregivers, which increases workload and chances of human error.
- Studies indicate that missed doses, incorrect timing, and wrong medication intake can lead to serious health complications.
- Automation in healthcare has emerged as a reliable solution to improve accuracy, safety, and efficiency in medication management.
- Automated medicine dispensing systems aim to ensure correct dosage at the right time while minimizing human involvement.

2. Review of Existing Medicine Dispensing Technologies

- Early medicine dispensers were mechanical and time-based, requiring manual refilling and setup.
- Research by R. Kumar et al. (2020) highlights that timer-based pill boxes reduce missed doses but lack remote monitoring.
- Some commercial systems use RFID and barcode scanning for medicine verification, but these systems are costly and complex.
- IoT-based medicine dispensers introduced remote monitoring but often depend on cloud servers and high-power processors.
- Literature suggests the need for a **low-cost, standalone, Wi-Fi-enabled solution** suitable for both hospitals and home environments.

3. Microcontroller Selection and Comparative Analysis

- The choice of controller plays a critical role in automation, connectivity, and system cost.
- Arduino Uno is widely used in basic automation but requires an external Wi-Fi module, increasing cost and complexity.
- Raspberry Pi offers high processing power but is expensive, consumes more power, and is unsuitable for simple control tasks.
- Research comparisons show that **ESP32** provides an optimal balance of performance, cost, and power efficiency.
- ESP32 integrates Wi-Fi, has a dual-core processor (up to 240 MHz), and sufficient GPIO pins, making it ideal for smart healthcare devices.

4. Actuation Mechanism Using Servo Motors

- Servo motors are commonly used in dispensing mechanisms due to their precise angular control.
- Studies show that servo-based dispensing ensures accurate release of medicines from predefined compartments.
- Compared to DC motors, servos eliminate the need for complex feedback mechanisms.
- Research by S. Patel et al. (2021) confirms that servo-controlled dispensers significantly reduce dosage errors.

5. Wireless Control and Web-Based Interfaces

- Wireless control improves accessibility and reduces physical interaction, which is crucial in hospital and ICU environments.
- Literature indicates that embedded web servers are more reliable than mobile applications for local control.
- ESP32's ability to host a web server enables real-time control without external cloud dependency.
- Research highlights that web-based interfaces simplify operation for non-technical users such as patients and caregivers.
- Wi-Fi-based control also enables remote monitoring, improving flexibility and response time.

6. Safety, Alert, and Notification Systems

- Alert mechanisms are essential to ensure patient awareness during medicine dispensing.
- Buzzer-based alerts are widely used due to simplicity and effectiveness.
- Studies show that audio alerts improve medication adherence among elderly patients.
- Combining alerts with automation reduces the dependency on caregivers.
- Literature supports integrating alarms with dispensing actions to ensure correct medicine intake.

7. Applications in Hospital and Home-Care Environments

- Automated medicine dispensers reduce nurse workload in hospitals by handling routine medicine administration.
- They minimize medication errors, especially in high-pressure environments such as emergency wards.
- In isolation wards, automated systems reduce physical contact, improving infection control.
- Home-care patients benefit from timely medication without constant supervision.
- Research emphasizes their importance in managing multiple patients efficiently.

8. Advantages and System Impact

- Improved medication accuracy and timing.
- Reduction in human error and caregiver dependency.
- Low-cost implementation using minimal components.
- User-friendly and portable design.
- Scalable architecture suitable for future enhancements such as mobile alerts or cloud logging.

9. Research Gaps and Future Scope

- Most existing systems are either expensive or dependent on cloud infrastructure.
- Limited focus on standalone, low-power, Wi-Fi-enabled devices.
- Future research suggests integrating mobile notifications, real-time dosage tracking, and AI-based adherence analysis.
- Expansion to include biometric authentication can further improve safety.
- Integration with hospital management systems can enhance large-scale deployment.

10. REFERENCES

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